

===== CORNER: ff_100C_1v65 =====

Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxt1_1)
0.2213	0.2213	^ _9167_/Q (sky130_fd_sc_hd_dfxt1_1)
0.0000	0.2213	^ _8135_/D (sky130_fd_sc_hd_dfxt1_1)
	0.2213	data arrival time
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxt1_1)
-0.0277	-0.0277	library hold time
	-0.0277	data required time
	-0.0277	data required time
	-0.2213	data arrival time
	0.2491	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxt1_1)
6.9895	6.9895	^ _9175_/Q (sky130_fd_sc_hd_dfxt1_1)
-0.5179	6.4716	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	6.4716	v _8540_/D (sky130_fd_sc_hd_dfxt1_1)
	6.4716	data arrival time
11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxt1_1)
-0.3431	10.6569	library setup time
	10.6569	data required time
	10.6569	data required time
	-6.4716	data arrival time
	4.1853	slack (MET)

wns max 0.00

tns max 0.00

worst slack max 4.19

worst slack min 0.25

===== CORNER: ff_100C_1v95 =====

Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxt1_1)
0.1768	0.1768	^ _9167_/Q (sky130_fd_sc_hd_dfxt1_1)
0.0000	0.1768	^ _8135_/D (sky130_fd_sc_hd_dfxt1_1)
	0.1768	data arrival time
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxt1_1)
-0.0192	-0.0192	library hold time
	-0.0192	data required time
	-0.0192	data required time
	-0.1768	data arrival time
	0.1960	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Delay	Time	Description
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Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (skyl130_fd_sc_hd_dfxt1)
5.8987	5.8987	^ _9175_/Q (skyl130_fd_sc_hd_dfxt1)
-0.8109	5.0878	v _6573_/Y (skyl130_fd_sc_hd_a2loi1)
0.0000	5.0878	v _8540_/D (skyl130_fd_sc_hd_dfxt1)
	5.0878	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (skyl130_fd_sc_hd_dfxt1)
-0.2849	10.7151	library setup time
	10.7151	data required time

	10.7151	data required time
	-5.0878	data arrival time

	5.6274	slack (MET)

wns max 0.00
tns max 0.00
worst slack max 5.63
worst slack min 0.20
===== CORNER: ff_n40C_1v56 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (skyl130_fd_sc_hd_dfxt1)
0.2625	0.2625	^ _9167_/Q (skyl130_fd_sc_hd_dfxt1)
0.0000	0.2625	^ _8135_/D (skyl130_fd_sc_hd_dfxt1)
	0.2625	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (skyl130_fd_sc_hd_dfxt1)
-0.0291	-0.0291	library hold time
	-0.0291	data required time

	-0.0291	data required time
	-0.2625	data arrival time

	0.2915	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (skyl130_fd_sc_hd_dfxt1)
9.0010	9.0010	^ _9175_/Q (skyl130_fd_sc_hd_dfxt1)
-0.4392	8.5618	v _6573_/Y (skyl130_fd_sc_hd_a2loi1)
0.0000	8.5618	v _8540_/D (skyl130_fd_sc_hd_dfxt1)
	8.5618	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (skyl130_fd_sc_hd_dfxt1)
-0.6335	10.3665	library setup time
	10.3665	data required time

	10.3665	data required time
	-8.5618	data arrival time

	1.8047	slack (MET)

wns max 0.00
tns max 0.00
worst slack max 1.80
worst slack min 0.29
===== CORNER: ff_n40C_1v65 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)

0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxt1)
0.2304	0.2304	^ _9167_/Q (sky130_fd_sc_hd_dfxt1)
0.0000	0.2304	^ _8135_/D (sky130_fd_sc_hd_dfxt1)
	0.2304	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxt1)
-0.0247	-0.0247	library hold time
	-0.0247	data required time

	-0.0247	data required time
	-0.2304	data arrival time

	0.2551	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxt1)
8.0709	8.0709	^ _9175_/Q (sky130_fd_sc_hd_dfxt1)
-0.7394	7.3316	v _6573_/Y (sky130_fd_sc_hd_a21oi1)
0.0000	7.3316	v _8540_/D (sky130_fd_sc_hd_dfxt1)
	7.3316	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxt1)
-0.4897	10.5103	library setup time
	10.5103	data required time

	10.5103	data required time
	-7.3316	data arrival time

	3.1788	slack (MET)

wns max 0.00
 tns max 0.00
 worst slack max 3.18
 worst slack min 0.26
 ===== CORNER: ff_n40C_1v76 =====
 Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxt1)
0.2026	0.2026	^ _9167_/Q (sky130_fd_sc_hd_dfxt1)
0.0000	0.2026	^ _8135_/D (sky130_fd_sc_hd_dfxt1)
	0.2026	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxt1)
-0.0218	-0.0218	library hold time
	-0.0218	data required time

	-0.0218	data required time
	-0.2026	data arrival time

	0.2243	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxt1)
7.3016	7.3016	^ _9175_/Q (sky130_fd_sc_hd_dfxt1)
-0.9247	6.3768	v _6573_/Y (sky130_fd_sc_hd_a21oi1)
0.0000	6.3768	v _8540_/D (sky130_fd_sc_hd_dfxt1)
	6.3768	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)

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0.0000    11.0000    clock reconvergence pessimism
11.0000 ^ _8540_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.3819   10.6181    library setup time
10.6181   10.6181    data required time
-----
10.6181   data required time
-6.3768   data arrival time
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4.2413    slack (MET)

wns max 0.00
tns max 0.00
worst slack max 4.24
worst slack min 0.22
===== CORNER: ff_n40C_1v95 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

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Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtp_1)
0.1699	0.1699	^ _9167_/Q (sky130_fd_sc_hd_dfxtp_1)
0.0000	0.1699	^ _8135_/D (sky130_fd_sc_hd_dfxtp_1)
	0.1699	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.0176	-0.0176	library hold time
	-0.0176	data required time

	-0.0176	data required time
	-0.1699	data arrival time

	0.1875	slack (MET)

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Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

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Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtp_1)
6.2945	6.2945	^ _9175_/Q (sky130_fd_sc_hd_dfxtp_1)
-1.1034	5.1911	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	5.1911	v _8540_/D (sky130_fd_sc_hd_dfxtp_1)
	5.1911	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.3002	10.6998	library setup time
	10.6998	data required time

	10.6998	data required time
	-5.1911	data arrival time

	5.5087	slack (MET)

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wns max 0.00
tns max 0.00
worst slack max 5.51
worst slack min 0.19
===== CORNER: ss_100C_1v40 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

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Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtp_1)
0.7516	0.7516	v _9167_/Q (sky130_fd_sc_hd_dfxtp_1)
0.0000	0.7516	v _8135_/D (sky130_fd_sc_hd_dfxtp_1)
	0.7516	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.1537	-0.1537	library hold time

-0.1537	data required time

-0.1537	data required time
-0.7516	data arrival time

0.9053	slack (MET)

Startpoint: _9060_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _9135_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9060_/CLK (sky130_fd_sc_hd_dfxtpl_1)
0.7725	0.7725	v _9060_/Q (sky130_fd_sc_hd_dfxtpl_1)
3.1277	3.9002	v _4020_/X (sky130_fd_sc_hd_or4_1)
1.2824	5.1827	^ _4021_/Y (sky130_fd_sc_hd_clkinv_1)
0.9296	6.1122	v _4163_/Y (sky130_fd_sc_hd_a3loi_1)
12.0133	18.1255	^ _4168_/Y (sky130_fd_sc_hd_a211loi_0)
2.2822	20.4077	v _4211_/Y (sky130_fd_sc_hd_a22loi_1)
1.5755	21.9832	v _4212_/X (sky130_fd_sc_hd_and3_1)
0.2209	22.2041	^ _4213_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	22.2041	^ _9135_/D (sky130_fd_sc_hd_dfxtpl_1)
	22.2041	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _9135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.3076	10.6924	library setup time
	10.6924	data required time

	10.6924	data required time
	-22.2041	data arrival time

	-11.5117	slack (VIOLATED)

wns max -11.51
 tns max -9421.76
 worst slack max -11.51
 worst slack min 0.91
 ===== CORNER: ss_100C_1v60 =====
 Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
0.5312	0.5312	v _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	0.5312	v _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	0.5312	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.1108	-0.1108	library hold time
	-0.1108	data required time

	-0.1108	data required time
	-0.5312	data arrival time

	0.6420	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
 Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
 Path Group: clk
 Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtpl_1)
13.4738	13.4738	^ _9175_/Q (sky130_fd_sc_hd_dfxtpl_1)
1.2540	14.7278	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	14.7278	v _8540_/D (sky130_fd_sc_hd_dfxtpl_1)
	14.7278	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-1.0764	9.9236	library setup time
	9.9236	data required time

9.9236	data required time	
-14.7278	data arrival time	

-4.8042	slack (VIOLATED)	

wns max -4.80
tns max -3412.85
worst slack max -4.80
worst slack min 0.64
===== CORNER: ss_n40C_1v28 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
1.4545	1.4545	v _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	1.4545	v _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	1.4545	data arrival time
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.3751	-0.3751	library hold time
	-0.3751	data required time

	-0.3751	data required time
	-1.4545	data arrival time

	1.8296	slack (MET)

Startpoint: _9060_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _9135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9060_/CLK (sky130_fd_sc_hd_dfxtpl_1)
1.4823	1.4823	v _9060_/Q (sky130_fd_sc_hd_dfxtpl_1)
6.9965	8.4788	v _4020_/X (sky130_fd_sc_hd_or4_1)
2.5467	11.0255	^ _4021_/Y (sky130_fd_sc_hd_clkinv_1)
1.5882	12.6137	v _4163_/Y (sky130_fd_sc_hd_a3loi_1)
33.1756	45.7893	^ _4168_/Y (sky130_fd_sc_hd_a211loi_0)
15.1059	60.8952	v _4211_/Y (sky130_fd_sc_hd_a22loi_1)
5.0997	65.9949	v _4212_/X (sky130_fd_sc_hd_and3_1)
0.5847	66.5796	^ _4213_/Y (sky130_fd_sc_hd_a21loi_1)
0.0000	66.5796	^ _9135_/D (sky130_fd_sc_hd_dfxtpl_1)
	66.5796	data arrival time
11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _9135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.5778	10.4222	library setup time
	10.4222	data required time

	10.4222	data required time
	-66.5796	data arrival time

	-56.1574	slack (VIOLATED)

wns max -56.16
tns max -49322.57
worst slack max -56.16
worst slack min 1.83
===== CORNER: ss_n40C_1v35 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
1.0809	1.0809	v _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	1.0809	v _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	1.0809	data arrival time
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism

	0.0000	^	_8135_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.2666	-0.2666		library hold time
	-0.2666		data required time

	-0.2666		data required time
	-1.0809		data arrival time

	1.3475		slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtp_1)
32.2639	32.2639	^ _9175_/Q (sky130_fd_sc_hd_dfxtp_1)
10.2862	42.5501	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	42.5501	v _8540_/D (sky130_fd_sc_hd_dfxtp_1)
	42.5501	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtp_1)
-3.6355	7.3645	library setup time
	7.3645	data required time

	7.3645	data required time
	-42.5501	data arrival time

	-35.1855	slack (VIOLATED)

wns max -35.19

tns max -30463.35

worst slack max -35.19

worst slack min 1.35

===== CORNER: ss_n40C_1v40 =====

Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtp_1)
0.9091	0.9091	v _9167_/Q (sky130_fd_sc_hd_dfxtp_1)
0.0000	0.9091	v _8135_/D (sky130_fd_sc_hd_dfxtp_1)
	0.9091	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.2158	-0.2158	library hold time
	-0.2158	data required time

	-0.2158	data required time
	-0.9091	data arrival time

	1.1249	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)

Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk

Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtp_1)
27.3668	27.3668	^ _9175_/Q (sky130_fd_sc_hd_dfxtp_1)
7.4930	34.8598	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	34.8598	v _8540_/D (sky130_fd_sc_hd_dfxtp_1)
	34.8598	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtp_1)
-3.2255	7.7745	library setup time
	7.7745	data required time

	7.7745	data required time
	-34.8598	data arrival time

-27.0853 slack (VIOLATED)

wns max -27.09
tns max -22870.21
worst slack max -27.09
worst slack min 1.12
===== CORNER: ss_n40C_1v44 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
0.8008	0.8008	v _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	0.8008	v _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	0.8008	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.1901	-0.1901	library hold time
	-0.1901	data required time

	-0.1901	data required time
	-0.8008	data arrival time

	0.9909	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtpl_1)
24.2716	24.2716	^ _9175_/Q (sky130_fd_sc_hd_dfxtpl_1)
6.8925	31.1641	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	31.1641	v _8540_/D (sky130_fd_sc_hd_dfxtpl_1)
	31.1641	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-2.5429	8.4571	library setup time
	8.4571	data required time

	8.4571	data required time
	-31.1641	data arrival time

	-22.7070	slack (VIOLATED)

wns max -22.71
tns max -18864.62
worst slack max -22.71
worst slack min 0.99
===== CORNER: ss_n40C_1v60 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
0.5413	0.5413	v _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	0.5413	v _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	0.5413	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.1215	-0.1215	library hold time
	-0.1215	data required time

	-0.1215	data required time
	-0.5413	data arrival time

	0.6628	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtp_1)
17.0297	17.0297	^ _9175_/Q (sky130_fd_sc_hd_dfxtp_1)
2.8454	19.8751	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	19.8751	v _8540_/D (sky130_fd_sc_hd_dfxtp_1)
	19.8751	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtp_1)
-1.4970	9.5030	library setup time
	9.5030	data required time

	9.5030	data required time
	-19.8751	data arrival time

	-10.3721	slack (VIOLATED)

wns max -10.37
tns max -7893.63
worst slack max -10.37
worst slack min 0.66
===== CORNER: ss_n40C_1v76 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtp_1)
0.4099	0.4099	v _9167_/Q (sky130_fd_sc_hd_dfxtp_1)
0.0000	0.4099	v _8135_/D (sky130_fd_sc_hd_dfxtp_1)
	0.4099	data arrival time

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtp_1)
-0.0939	-0.0939	library hold time
	-0.0939	data required time

	-0.0939	data required time
	-0.4099	data arrival time

	0.5038	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description

0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtp_1)
13.0289	13.0289	^ _9175_/Q (sky130_fd_sc_hd_dfxtp_1)
1.9177	14.9466	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	14.9466	v _8540_/D (sky130_fd_sc_hd_dfxtp_1)
	14.9466	data arrival time

11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtp_1)
-1.3188	9.6812	library setup time
	9.6812	data required time

	9.6812	data required time
	-14.9466	data arrival time

	-5.2654	slack (VIOLATED)

wns max -5.27
tns max -3417.96
worst slack max -5.27
worst slack min 0.50
===== CORNER: tt_025C_1v80 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)

Path Group: clk
Path Type: min

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
0.2749	0.2749	^ _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	0.2749	^ _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	0.2749	data arrival time
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.0346	-0.0346	library hold time
	-0.0346	data required time
	-0.0346	data required time
	-0.2749	data arrival time
	0.3096	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtpl_1)
8.4347	8.4347	^ _9175_/Q (sky130_fd_sc_hd_dfxtpl_1)
-0.2494	8.1853	v _6573_/Y (sky130_fd_sc_hd_a2loi_1)
0.0000	8.1853	v _8540_/D (sky130_fd_sc_hd_dfxtpl_1)
	8.1853	data arrival time
11.0000	11.0000	clock clk (rise edge)
0.0000	11.0000	clock network delay (ideal)
0.0000	11.0000	clock reconvergence pessimism
	11.0000	^ _8540_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.5544	10.4456	library setup time
	10.4456	data required time
	10.4456	data required time
	-8.1853	data arrival time
	2.2603	slack (MET)

wns max 0.00
tns max 0.00
worst slack max 2.26
worst slack min 0.31
==== CORNER: tt_100C_1v80 =====
Startpoint: _9167_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8135_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: min

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9167_/CLK (sky130_fd_sc_hd_dfxtpl_1)
0.2776	0.2776	^ _9167_/Q (sky130_fd_sc_hd_dfxtpl_1)
0.0000	0.2776	^ _8135_/D (sky130_fd_sc_hd_dfxtpl_1)
	0.2776	data arrival time
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	clock reconvergence pessimism
	0.0000	^ _8135_/CLK (sky130_fd_sc_hd_dfxtpl_1)
-0.0369	-0.0369	library hold time
	-0.0369	data required time
	-0.0369	data required time
	-0.2776	data arrival time
	0.3145	slack (MET)

Startpoint: _9175_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: _8540_ (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

Delay	Time	Description
0.0000	0.0000	clock clk (rise edge)
0.0000	0.0000	clock network delay (ideal)
0.0000	0.0000	^ _9175_/CLK (sky130_fd_sc_hd_dfxtpl_1)

7.9393	7.9393	^	_9175_/Q (sky130_fd_sc_hd__dfxtp_1)
-0.1592	7.7801	v	_6573_/Y (sky130_fd_sc_hd__a2loi_1)
0.0000	7.7801	v	_8540_/D (sky130_fd_sc_hd__dfxtp_1)
	7.7801		data arrival time
11.0000	11.0000		clock clk (rise edge)
0.0000	11.0000		clock network delay (ideal)
0.0000	11.0000		clock reconvergence pessimism
	11.0000	^	_8540_/CLK (sky130_fd_sc_hd__dfxtp_1)
-0.4755	10.5245		library setup time
	10.5245		data required time

	10.5245		data required time
	-7.7801		data arrival time

	2.7444		slack (MET)

wns max 0.00
 tns max 0.00
 worst slack max 2.74
 worst slack min 0.31